Theodore Schwartz

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Education

Brown University Providence, RI

Sc.B in Applied Mathematics-Computer Science, GPA: 4.0/4.0 *August 2025 – May 2029*

- **Relevant Coursework:** Computer Systems, Data Structures & Algorithms, Discrete Structures & Probability, Linear Algebra, Multivariable Calculus, Principles of Economics, Organic Chemistry I & II

Software and Research Experience

NYU Langone Health Grossman School of Medicine New York, NY

Research Assistant, Lang Lab - Neuroscience Institute *May 2026 – Present*

- Collaborate directly with the principal investigator to develop a Python, PyTorch, and scikit-image pipeline for large-scale analysis of animal cerebellum histology, enabling analyses previously impractical at dataset scale.
- Partner with a postdoctoral researcher to engineer behavioral research spaces and experimental rigs.
- Introduced laser cutting and 3D printing to prototype lower-cost laboratory components and streamlined daily feeding and testing workflows, reducing routine work by approximately 2 hours per day.

Vidyasirimedhi Institute of Science and Technology (VISTEC) Remote

Student Researcher / Research Assistant *September 2022 – June 2025*

- Designed reinforcement-learning and imitation-learning methods for human-AI cooperation using PyTorch; adapted training pipelines, generated offline datasets, and evaluated cooperative agents.
- Developed a lightweight diversity metric and validation method for cooperative RL agent populations.
- Authored two first-author papers; presented through ISEF, STS, and JSHS and submitted to NeurIPS.

Work and Volunteer Experience

Full Stack @ Brown Providence, RI

Developer *September 2025 – Present*

- Developed frontend and backend features for 3 team-built web apps using React, JavaScript, Firebase, and Git.
- Collaborated across dev teams through scoped feature ownership, code review, docs, and Git-based workflows.

Mental Health Hotline Providence, RI

Volunteer (Purposefully vague organization name for safety purposes) *September 2025-Present*

- Committed to 100+ on-call hours supporting people experiencing mental-health crises and suicidal ideation.
- Trained in active listening, crisis communication, risk assessment, and de-escalation in high-stakes situations.

Discover Camp Hawthorn, NY

Counselor *June 2023 – August 2025*

- Taught 25+ children aged 7-13 basic coding in Python, CAD skills, and other STEM-related activities.
- Managed safety protocols in high-stress environments involving laser cutters, 3D printers, and other equipment.

Leadership Experience

BearBots Robotics Team Briarcliff Manor, NY

Co-Founder & Co-President *September 2022 – June 2025*

- Co-founded an FTC robotics program where none existed, growing the organization from 2-3 students to 30.
- Designed software and hardware for 2 competition robots and contributed to strategy and fundraising.
- Trained 10+ leads to manage subteams, transforming a founder-dependent club into a sustainable organization.

Brown Bears Quadball Providence, RI

Communications Officer *September 2025 – Present*

- Lead recruitment, branding, and campus outreach initiatives to strengthen visibility and attract new players.
- Reorganized recruitment strategy, generating 1M+ Instagram views and attracting 100+ prospective players.

Selected Honors and Skills

Honors: National Merit Semifinalist | RIT Computing Award | Somers Science Fair - 1st, Computer Science & Mathematics | FTC THINK Award | FTC Design Award
Technical: Python (PyTorch, scikit-image), Java, JavaScript (React, Firebase), C++, C, Git, CAD
Additional: EMT, Spanish, Mental Health Crisis Management